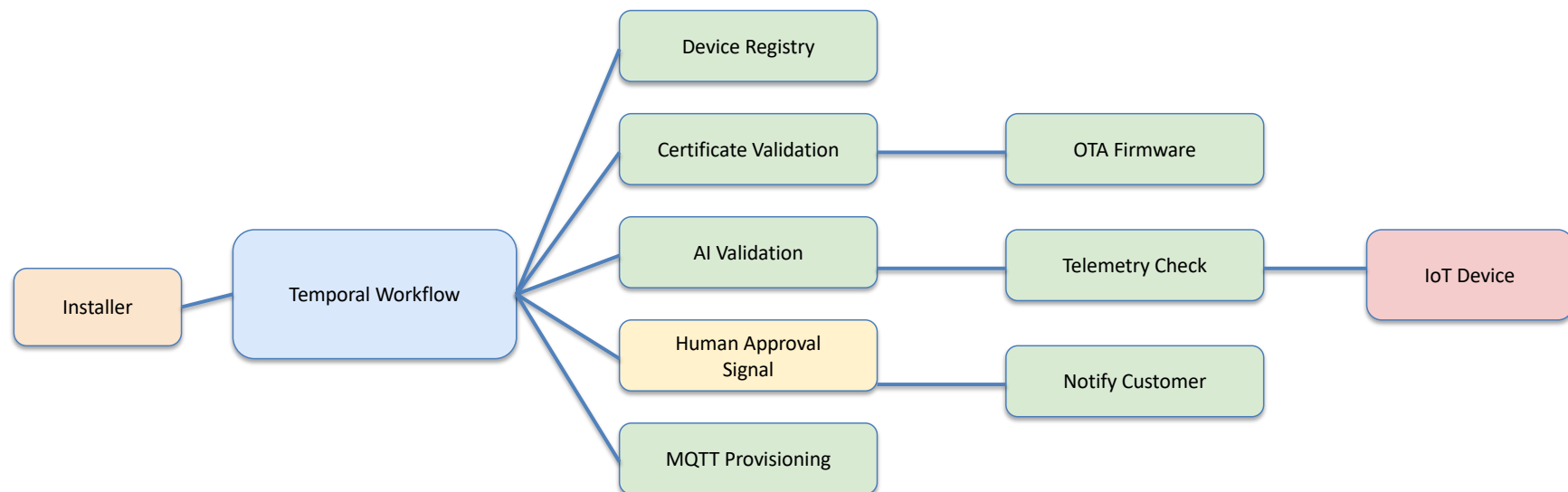


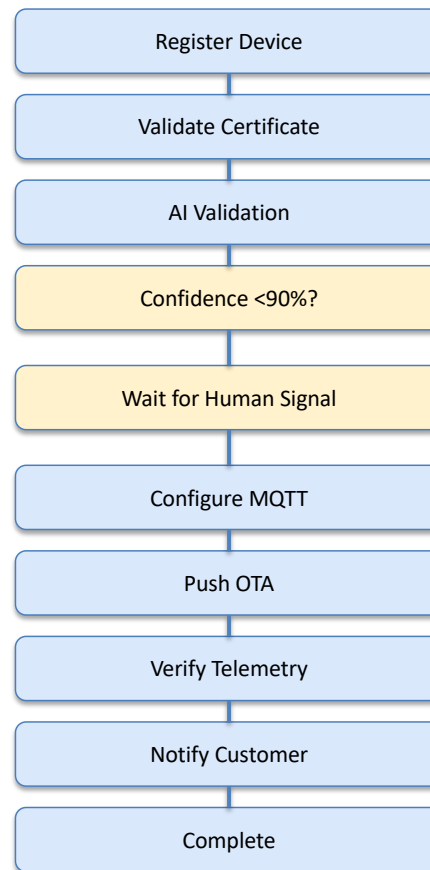
1. Business Problem

- Customers self-install IoT sensors across many sites.
- Onboarding spans multiple systems: registry, certificates, AI validation, MQTT, OTA, telemetry.
- Human approval may be required for low-confidence AI results.
- Failures should resume safely without duplicate provisioning or lost state.
- Goal: Orchestrate the lifecycle using Temporal durable workflows.

2. High-Level Architecture

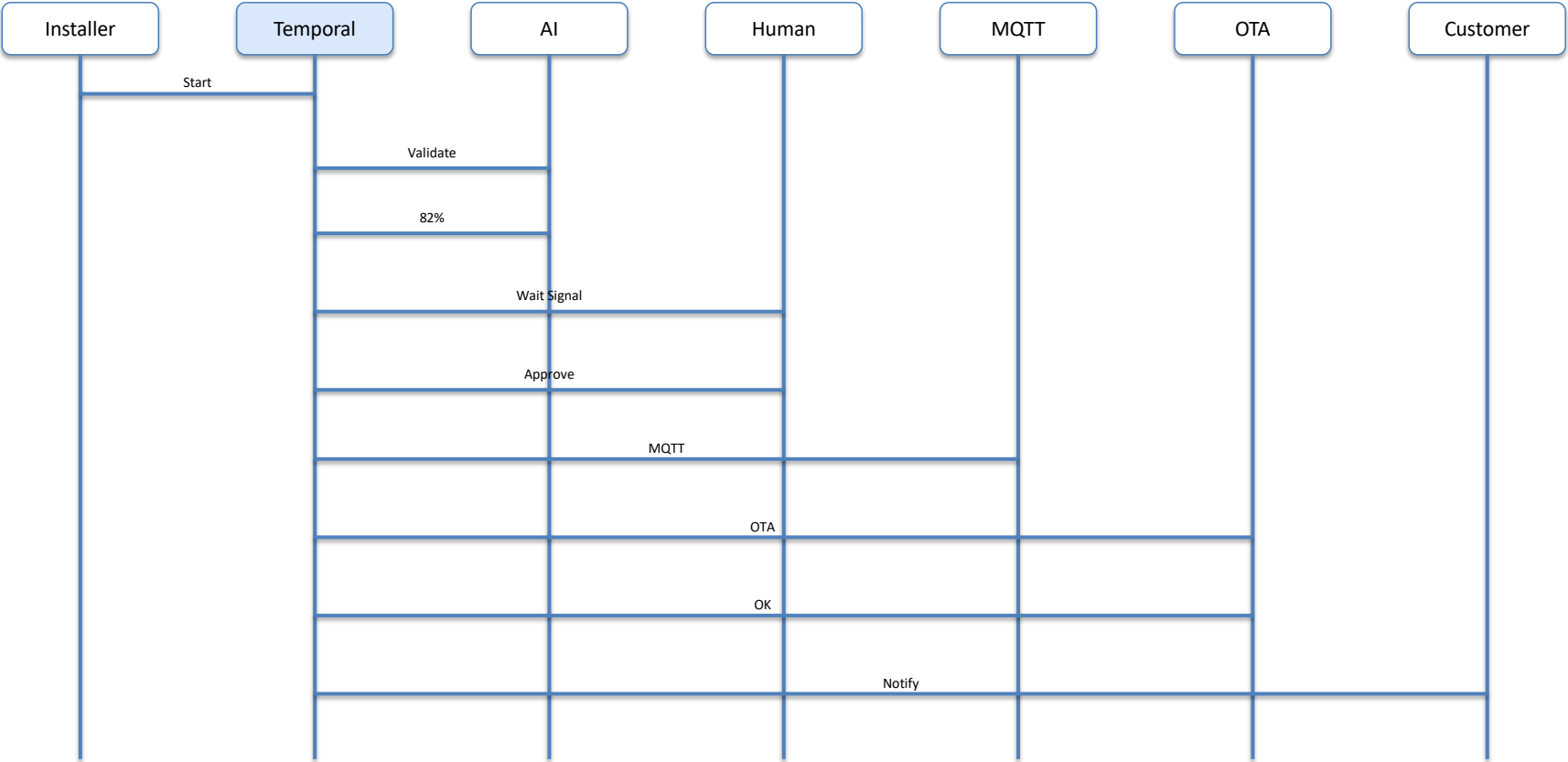


3. Workflow State Machine



- Retries
- Timers
- Signals
- Compensation
- Durable State

4. Sequence Diagram



5. Why Temporal?

- Durable execution preserves workflow state across failures.
- Activities encapsulate external system calls with automatic retries.
- Signals enable human-in-the-loop approvals without polling.
- Queries expose real-time workflow status to portals and operators.
- Built-in timers, compensation, and observability reduce orchestration complexity.
- Compared with queues or Step Functions, Temporal provides code-first, long-running workflow orchestration.